

Time : Three Hours]

[Maximum Marks : 50

Note : The candidates are required to attempt *three* questions each from Section-A and Section-B carrying 5 marks each and the entire Section-C consisting of 10 short questions carrying 2 marks each.

SECTION – A

- I. Define the Thevenin theorem. Explain steps used to solve Thevenin theorem.
- II. A coil of resistance 10 ohms and inductance 0.1 H is connected in series with a condenser of capacitance 150 micro farad across a 200 V, 50 Hz supply. Determine
 - (i) Impedance.
 - (ii) Current.
 - (iii) Power factor.
 - (iv) Voltage across coil.
 - (v) Voltage across condenser.
- III. Discuss the behaviour of RLC series circuit when applied with ac voltage with the help of phasor diagram. Also, derive the expression for series resonance.

32506-MJ/500/HHH/442

123 [P.T.O.]

- IV. Convert star connected set of 3 resistors R into delta connection.
- V. Using Two wattmeter method find power measurement in 3-phase AC system.

SECTION-B

- VI. Derive an expression for torque developed by a DC motor.
- VII. Explain the characteristics of DC series motor. Also explain its application.
- VIII. Define the role of commutator. Explain its working.
- IX. How open circuit test is performed on the transformer? Draw its set up and discuss the results obtained from the test.
- X. Explain principle, construction and working of transformer.

SECTION-C

- XI. Answer the following in short :
 - (a) Explain efficiency of transformer.
 - (b) Define Average value of A.C.
 - (c) Write the emf equation of transformer.
 - (d) Draw the phaser diagram for capacitor connected in series with ac supply.
 - (e) What is the working principle of D.C. motor?

- (f) Define Q-factor in series resonance.
 - (g) Define Peak Factor.
 - (h) Write down the relation between line current and phase current in star.
 - (i) State and explain Kirchhoff's current law.
 - (j) Define Transformation Ratio.
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